Lab - Research a Hardware Upgrade

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for two different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston HyperX Fury 8 GB DDR3 Ram 1600mhz CL10 PC3-12800U	Automatic overclocking up to 1866MHz Voltages start as low as 1.35V Asymmetric, aggressive heat spreader design Compatible with Intel 6, 7, 8, 9 and 1002 series chipsets, as well as A75, A87, A88, A89, A78 and E35 AMD chipsets	₱852
HKC 8 gb dd3 RAM 1600mhz PC3-12800	**Capacity**: 8GB - ideal for modern operating systems and multitasking **Type**: DDR3 SDRAM - widely compatible with systems from 2009-2016 **Speed**: 1600MHz (PC3-12800) - optimal performance for DDR3 systems **Reliability**: Quality tested for stable operation and longevity **Compatibility**: Works with most DDR3-compatible motherboards and laptops **Easy Installation**: Simple plug-and-play upgrade	₱849

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

For me the ram i would choose is the kingston hyperx as it is more reliable and runs better than HKC. HKC ddr3 used to be for budget option but since the hyperx fury is at the same price point then the kingston is a better choice for budget and quality ram.

© 2015 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public

Step 2: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
WD Red Pro 3TB NAS Hard Disk Drive - 7200 RPM SATA 6 Gb/s 64MB Cache 3.5 Inch - WD3001FFSX	• 7200 RPM spindle speed • 64MB cache • SATA 6 Gb/s interface • delivering up to 168 MB/s of reliable data transfer performance	₱5,449
Seagate Barracuda 3TB 7200rpm SATA III Internal Hard Drive (ST3000DM001)	• Capacity: 3 TB • Interface: SATA 3.0 Gb/s • Cache: 16 MB • Rotational Speed: 7200 rpm • Average Latency: 4.17msec	₱7,122
Ramsta R900 M.2 NVMe SSD 512GB R1R900512G	• Interface: SATA III 6Gb/s • Operating temperature: 0°C~70°C • Storage temperature: -40°C~80°C	₱6,699
Ramsta SSD S800 512GB S800-2.5-512G	• Capacity: 512GB • Product Size: 100x70x6.9mm • Interface：SATA III 6Gb/s • Operating temperature：0°C~70°C • Storage temperature：-40°C~85°C	₱3,749
SanDisk SSD PLUS 1000GB SDSSDA-1T00-G27	• Capacity: 1TB • Form Factor: 2.5-Inch • Sequential Read Performance: 535MB/s • Sequential Write: Performance: 350MB/s • Warranty: 3-Year Limited Warranty • Operating Temperature: 0°C to 70°C	₱4,750

Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

For HDD I would choose the WD Red Pro as it has longer lifespan, better NAS Durability, encrypted portable storage when compared to the Seagate HDD. As for SSD, personally I would choose 2 SSD to use, 512GB NVMe will be my C drive or boot drive as it is 5-10 times faster than SATA ssd, and I would choose the 1TB SATA SSD as my extra storage.

© 2015 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public

Step 3: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for two different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
Nvidia GeForce GTX 1060 (6GB) GP106-400-A1	Base Clock: 1506 MHz Boost Clock: 1709 MHz Memory Clock: 2002 MHz 8 Gbps effective Memory Size: 6 GB Memory Type: GDDR5 Memory Bus: 192 bit Bandwidth: 192.2 GB/s	₱4,935.12
GTX 1660 Ti TU116-400-A1	Base Clock: 1500 MHz Boost Clock: 1770 MHz Memory Clock: 1500 MHz 12 Gbps effective Memory Size: 6 GB Memory Type: GDDR6 Memory Bus: 192 bit Bandwidth: 288.0 GB/s	₱11,042.33

Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

For GPU, I think that the 1660 Ti is a better choice. As it is faster and stronger than the 1060. Although the 1060 is also a great option when it comes to lower budget, as it is still good even to this day and you can even play games on it at low to medium settings depending on the game.

© 2015 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public